

HARSH TAKALKAR

B.Tech Student – Electronics & Telecommunication Engineering

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B.Tech ECE student with hands-on experience training supervised ML models on real prediction problems, including an award-winning predictive maintenance system and a loan-risk classification model. Comfortable across the ML workflow — data cleaning, feature engineering, evaluation — using Python, scikit-learn, and Pandas, with strong signal-processing fundamentals.

EDUCATION

B.Tech, Electronics & Telecommunication Engineering

2025 – Present

SCTR's Pune Institute of Computer Technology (PICT), Pune – CGPA: 8.64

Diploma, Electronics & Telecommunication Engineering

2022 – 2025

Cusrow Wadia Institute of Technology, Pune – 94.82%, Diploma Topper

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

Machine Learning: NumPy, Pandas, Matplotlib, scikit-learn, Supervised & Unsupervised Learning, Data Analysis

Web Development: HTML, CSS, JavaScript

Tools & Platforms: Git, GitHub, Arduino, PCB Design, Vercel

EXPERIENCE

Electronics & Circuit Design Intern, Yog Electro – Chakan, Pune

6 weeks

- Assembled and tested electronic circuits and components in a manufacturing environment, validating functionality against design specifications
- Debugged hardware faults across production units, applying structured root-cause analysis to reduce rework

PROJECTS

DigiTwin — AI-Based Predictive Maintenance System

1st Prize, UpXGen Hackathon

- Built a digital twin using machine learning models to simulate real-time equipment behavior and predict failures before occurrence
- Designed an anomaly detection pipeline over streaming sensor data, cutting unplanned downtime by flagging deviations early; won 1st prize at UpXGen Hackathon

Technologies: Python, Machine Learning, Digital Twin, IoT

Human Presence Detection via Wi-Fi Signal Disturbance

- Trained a machine learning model to detect human presence from Wi-Fi channel signal disturbances, applying signal processing to extract features from noisy wireless data

Technologies: Python, Machine Learning, Signal Processing

Notes Sharing Platform

- Developed a full-stack web platform with a SQL backend for students to upload, browse, and share study notes organized by subject and year

Technologies: HTML, CSS, JavaScript, SQL

CreditWise — ML-Based Loan Approval System

- Trained supervised learning models in scikit-learn on applicant income, credit score, DTI ratio, and collateral data to classify loan applications as approved or rejected
- Engineered features across 19 applicant attributes to reduce manual bias in credit decisions, preparing the model for deployment as an automated pre-screening layer ahead of human verification

Technologies: Python, scikit-learn, Pandas, Machine Learning

Military Surveillance Vehicle & Launcher

- Engineered a remotely operated surveillance vehicle with Wi-Fi control and radar-based target detection

Technologies: Arduino, Wi-Fi Module, Radar, Embedded C